

Vanishing weak gradient tested by divergence

MathlibAnnex.WeakDivergenceZero

def

Expresses a weak equation through compactly supported continuously differentiable vector fields.

Statement

Let E be a finite-dimensional real normed vector space equipped with a measurable structure, let μ be any measure on E , let $U \subseteq E$, and let $u : E \rightarrow \mathbb{R}$. For a C^1 vector field $W : E \rightarrow E$, write $DW(z)$ for its real Fréchet derivative and $\operatorname{div} W(z) = \operatorname{tr}(DW(z))$. The weak equation considered here is

$$\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$$

for every C^1 vector field $W : E \rightarrow E$ supplied with a compact carrier $K \subseteq U$ containing its nonzero set. The integral is with respect to $\mu|_U$.

Definition

Using $\operatorname{div} W = \operatorname{tr}(DW)$, the complete condition is

$$\forall K \subseteq E \text{ compact}, \quad \forall W \in C^1(E, E), \quad \{z : W(z) \neq 0\} \subseteq K \subseteq U \implies \int_U u(z) \operatorname{div} W(z) d\mu(z) = 0.$$

The source packages K and W together as a compact test field. Without integrability hypotheses, the integral has the total Bochner-integral convention; no distribution-theoretic regularity is silently added.

Assumptions

The predicate itself assumes no openness of U , compatibility between the topology and measurable structure, Haar property of μ , or local integrability of u . These are separate hypotheses in its analytic uses. The zero-dimensional case is included.

Conclusion

The result of the definition is a proposition about u , U , and μ . It asks for the displayed equality for every admissible pair (K, W) , rather than for a derivative of u . In the later Haar setting, with U open and u locally integrable on U , the compact support and continuous bounded divergence make these test integrals ordinary integrable pairings.

Notes

A carrier is a chosen compact containing $\{z : W(z) \neq 0\}$; it need not equal that set or its closure. Since E is Hausdorff, K is closed and also contains $\overline{\{z : W(z) \neq 0\}}$. If $z \notin K$, the field is zero on the open neighborhood $E \setminus K$, hence $DW(z) = 0$ and $\operatorname{div} W(z) = 0$. This neighborhood argument, not merely $W(z) = 0$ at one point, confines the divergence to K .

Main citations

- Exact declaration and its source — `MathlibAnnex.WeakDivergenceZero`
- Trace definition of divergence — `MathlibAnnex.divergence`
- Divergence vanishes off the chosen carrier —
`MathlibAnnex.CompactC1VectorField.divergence_eq_zero_of_not_mem_carrier`
- Nonzero divergence lies in the carrier —
`MathlibAnnex.CompactC1VectorField.support_divergence_subset`
- A compact carrier paired with a supported C^1 vector field — `MathlibAnnex.CompactC1VectorField`
- The chosen carrier projection — `MathlibAnnex.CompactC1VectorField.carrier`
- Compactness of the carrier — `MathlibAnnex.CompactC1VectorField.isCompact_carrier`
- The nonzero set lies in the carrier — `MathlibAnnex.CompactC1VectorField.support_subset`
- The closed support lies in the carrier — `MathlibAnnex.CompactC1VectorField.tsupport_subset`
- Continuous differentiability of a test field — `MathlibAnnex.CompactC1VectorField.contDiff`
- Continuity of a test field — `MathlibAnnex.CompactC1VectorField.continuous`
- Compact support of a test field — `MathlibAnnex.CompactC1VectorField.hasCompactSupport`
- Packaging a C^1 field with a compact carrier — `MathlibAnnex.CompactC1VectorField.ofSupport`
- Evaluation of the packaged field — `MathlibAnnex.CompactC1VectorField.ofSupport_apply`
- Carrier of the packaged field — `MathlibAnnex.CompactC1VectorField.carrier_ofSupport`

Lean source signature (exact)

```
def WeakDivergenceZero [FiniteDimensional ℝ E]
  (μ : Measure E) (U : Set E) (u : E → ℝ) : Prop :=
  ∀ W : CompactC1VectorField E, W.carrier ⊆ U →
    ∫ y in U, u y * divergence W y ∂μ = 0
```

In the source	Mathematical meaning
<code>{E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]</code> <code>[MeasurableSpace E] [FiniteDimensional ℝ E]</code>	A finite-dimensional real normed space with any measurable structure.
<code>(μ : Measure E) (U : Set E) (u : E → ℝ) : Prop</code>	An arbitrary measure μ , subset U and scalar function u . The output is a proposition, with no openness, Haar or local-integrability assumption.
<code>∀ W : CompactC1VectorField E</code>	Every compact test-field pair: a compact set K and a C^1 vector field $W : E \rightarrow E$ whose nonzero set lies in K . The cited type is $\Sigma K : \text{Compacts } E, \text{ContDiffMapSupportedIn } E \text{ } E \text{ } 1 \text{ } K$.

In the source	Mathematical meaning
$W.\text{carrier} \subseteq U$	The chosen carrier K must lie in U . It contains the closed support of W but need not equal that support.
$\int y \text{ in } U, u \ y * \text{divergence } W \ y \ \partial\mu = 0$	The defining condition is $\int_U u(y) \operatorname{tr}(DW(y)) \, d\mu(y) = 0$, using $\mu _U$. With the preceding universal quantifier and implication this is the full RHS; the total integral alone does not certify integrability.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 5–9:

```

noncomputable section
open Set MeasureTheory
namespace MathlibAnnex
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
[MeasurableSpace E]

```

▼ Exact content identity

Declaration: `MathlibAnnex.WeakDivergenceZero`

Accepted content SHA-256: `ccdf776449c3dbc9c6659797735ca2313bb61789dfe48eed40962dfed471a60c`

Accepted source guide SHA-256: `08563edf1c679f0f913bf0d903821ed154a1849e372d0de93f2e1b2e2eef27f7`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`